530.334 Heat Transfer – Spring 2026

Computer Project 1

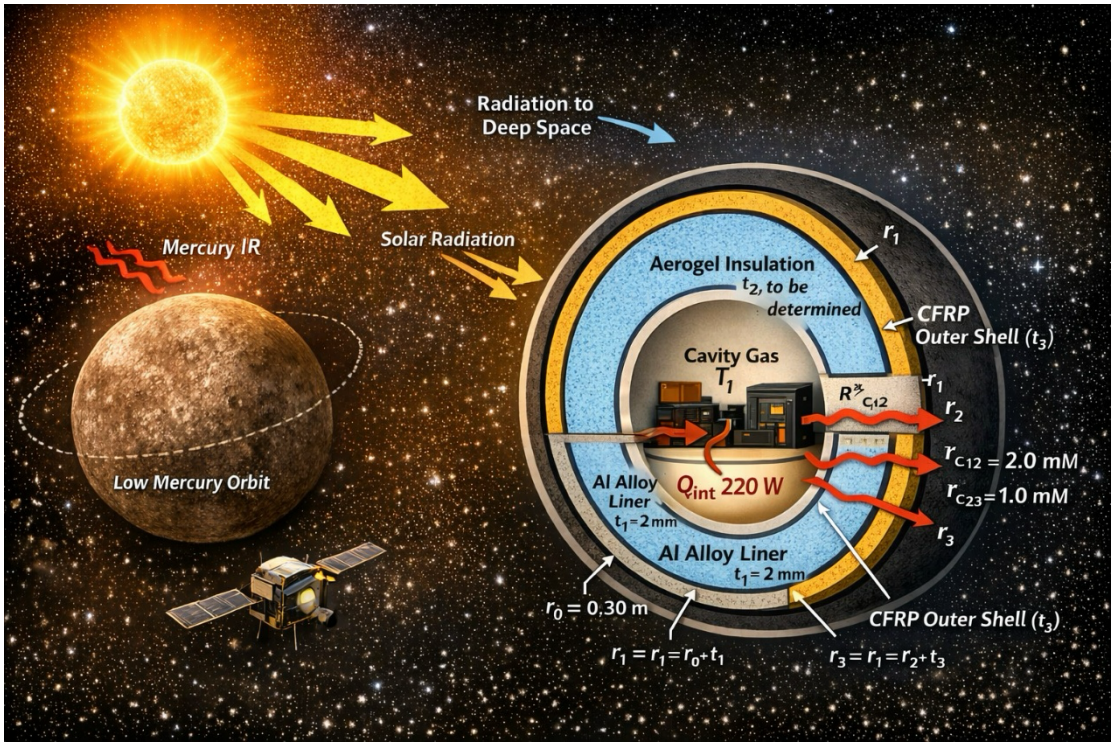
Due date: Gradescope, 3/6/26 10am

Thermal insulation/heat dissipation design for satellites orbiting Mercury (steady-state, 1D radial, nonlinear)

Background

A small scientific satellite orbits Mercury in an environment close to a vacuum. Its outer surface is heated by both shortwave solar radiation and infrared radiation from Mercury, and dissipates heat through radiation into deep space. Internal electronic equipment generates steady-state waste heat, which is transferred to the inner walls via natural convection of gas within the satellite's cavity, and then to the outer surface through radial heat conduction (including interfacial contact thermal resistance) of the multi-layered shell, and finally dissipated outwards through radiation.

For preliminary engineering estimation, the satellite is simplified as a concentric spherical shell composite structure, and a steady-state 1D radial model is established.



Geometry (Concentric 3-layer spherical shell)

Assume temperature varies only with radius:

$$T = T(r)$$

Inner cavity radius:

$$r_0 = 0.30 \text{ m}$$

Three solid layers (inside → outside):

Layer 1: Al alloy liner — thickness $t_1 = 2$ mm, region $r_0 \to r_1$

Layer 2: Aerogel insulation — thickness t_2(to be determined), region $r_1 \to r_2$

Layer 3: CFRP outer shell — thickness $t_3 = 3$ mm, region $r_2 \to r_3$

Design variable range:

$$1 \text{ mm} \leq t_2 \leq 40 \text{ mm}$$

Area-specific contact resistances:

At $r = r_1$(Layer 1 / Layer 2):

$$R''_{c,12} = 2.0 \times 10^{-4} m^2 K/W$$

At $r = r_2$(Layer 2 / Layer 3):

$$R''_{c,23} = 1.0 \times 10^{-4} m^2 K/W$$

2. Internal Heat Source + Internal Natural Convection (nonlinear h_i)

Electronics steady waste heat:

$$Q_{\text{int}} = 50 \text{ W}$$

Cavity gas temperature is uniform: T_g(unknown).

Natural convection correlation (simplified enclosure form):

$$Ra = \frac{g_{\text{eff}} \beta \mid T_g - T_w \mid L_c^3}{\nu \alpha}$$

$$Nu = 2 + 0.55\, Ra^{1/4}$$

$$h_i(T_g, T_w) = \frac{Nu\, k_g}{L_c}$$

Given gas properties (assume constants):

- $k_g = 0.030 \text{ W/(m} \cdot \text{K)}$
- $\nu = 1.7 \times 10^{-5} \text{ m}^2/\text{s}$
- $\alpha = 2.4 \times 10^{-5} \text{ m}^2/\text{s}$
- $\beta = 1/300 \text{ K}^{-1}$

- characteristic length $L_c = r_0$
- effective disturbance gravity $g_{\text{eff}} = 0.02\ \text{m/s}^2$

3.Conduction in Solid Layers (steady 1D spherical, with $k(T)$)

Temperature-dependent conductivities (must be used):

- **Layer 1 (Al):**

$$k_1(T) = 160[1 - 2.0 \times 10^{-4}(T - 300)]$$

- **Layer 2 (Aerogel):**

$$k_2(T) = 0.035 + 2.0 \times 10^{-4}(T - 300) + 1.0 \times 10^{-7}(T - 300)^2$$

- **Layer 3 (CFRP):**

$$k_3(T) = 6.0[1 + 1.0 \times 10^{-4}(T - 300)]$$

4.Outer Surface Radiation Boundary (solar + Mercury IR + radiation to space)

Deep-space background temperature:

$$T_{\text{space}} = 3\ \text{K}$$

Uniform-equivalent solar irradiance at Mercury orbit and geometric factor:

$$G_{\odot} = 9100\, W/m^2$$
$$g_{\odot} = 0.25$$

Mercury IR equivalent irradiance and geometric factor:

$$G_M = 1200\, W/m^2$$

$$g_m = 0.25$$

Outer surface temperature: $T_s = T(r_3)$.

Optical properties vary with temperature

$$\alpha(T) = 0.10 + 1.5 \times 10^{-4}(T - 300)$$
$$\varepsilon(T) = 0.92 - 1.0 \times 10^{-4}(T - 300)$$

5.Design Requirements

a. Electronics environment limit (represented by cavity gas temperature):

$$T_g \leq 360 \text{ K}$$

b. Outer surface temperature limit:

$$T_s \leq 450 \text{ K}$$

Tasks

(A) Model Formulation

Write a complete, solvable steady-state model that includes:

1. Governing equation for 1D radial conduction in each layer with $k(T)$.
2. Interface conditions at $r = r_1$and $r = r_2$: heat flow continuity and temperature jumps due to contact resistances.
3. Inner boundary at $r = r_0$: convection coupling with $h_i(T_g, T_w)$and the cavity energy balance.
4. Outer boundary at $r = r_3$: nonlinear radiation balance including T^4, $\alpha(T_s)$, and $\varepsilon(T_s)$.

Submit: your (i) equation set (with a clear unknown list) , (ii) a brief description of your numerical solution approach, (iii) a short paragraph explaining which resistance is expected to dominate and why?.

(B) Baseline Case: $t_2 = 10$ mm

Solve for the steady-state and report:

1. T_g, $T_w = T(r_0)$, $T_s = T(r_3)$, and total heat flow Q
2. Interface temperature jumps:

$$\Delta T_{c,12} = T(r_1^+) - T(r_1^-), \Delta T_{c,23} = T(r_2^+) - T(r_2^-)$$

Determine whether both design requirements are satisfied.

Submit: the numerical values (with units) and your pass/fail conclusion.

(C) Maximum Insulation Thickness t_2

Sweep $t_2 \in [1,40]$ mmand determine the minimum $t_2$such that:

$$T_g \leq 360, T_s \leq 450$$

Submit:

1. Plot $T_g(t_2)$with a line at 360 K
2. Plot $T_s(t_2)$with a line at 450 K
3. The maximum t_2 value

(D) Radial Temperature Distribution (use the maximum $t_2$from C)

Plot the temperature distribution $T(r)$from r_0 to $r_3$and clearly indicate the three layers. Also report values near each interface to show the temperature discontinuities caused by contact resistances.

Submit: $T(r)$ plot and the two interface jump magnitudes (K).

(E) Sensitivity Studies (fix $t_2 = t_{2max}$)

Electronics power sensitivity: Sweep $Q_{\text{int}} = 20\text{–}200$ W. Plot $T_g(Q_{\text{int}})$and determine the maximum allowable power $Q_{\text{int,max}}$such that $T_g = 360$ K.
Submit: both plots and the value of $Q_{\text{int,max}}$.

Notes (Allowed / Recommended)

1. You may discretize the radial domain using a simple finite-difference or finite-volume method (e.g., 10–30 nodes per layer).
2. Nonlinearity arises from T^4, $k(T)$, $\alpha(T)$, $\varepsilon(T)$, and $h_i(T_g, T_w)$. Use an iterative method (Picard) or a nonlinear solver (fsolve).
3. State your convergence criterion (e.g., max temperature change < 10^{-4} Kor energy imbalance < 0.1%).

Required Materials for Submission*. You are expected to turn in the plots as well as the code for all parts on Gradescope in one fine styled report. Please submit any written work as well. Be sure that you include comments throughout your code.*

Reminder: *All submitted materials should be your original work. Taking someone else's code (including your classmates, friends, Chat GPT, or other AI algorithms) and submitting it as if it is your own work will be considered plagiarism and will NOT be treated lightly. If you have any questions about what does or does not constitute plagiarism, please ask the course instructor or teaching assistants. Good luck!*